

Section 2.5: Physical Applications

Definition 14.

$$mass = (density) \cdot (volume)$$

Definition 15. Mass - One-dimensional object. A thin bar resting on $a \leq x \leq b$ with density ρ has:

$$mass = \int_a^b \rho(x) \, dx$$

Example 102. A thin two-meter bar made of an alloy with density in units of $\frac{kg}{m}$ is given by $\rho(x) = 1 + x^2$, for $x \geq 0$. Find the mass of the bar.

Definition 16. Mass - Circular object with radial density. For a two-dimensional disk with radius R and uniform radial density $\rho(x)$, for $0 \leq x \leq R$, the mass is measured by:

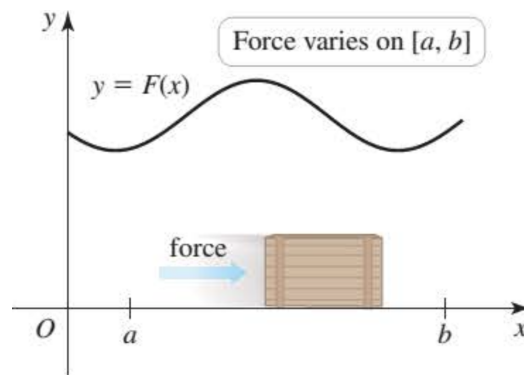
$$mass = \int_0^R 2\pi \cdot x \cdot \rho(x) \, dx$$

Example 103. Consider a plate with a radius of 8 inches and a radial density given by $\rho(x) = 2 + 5x^{-2/3}$. What is the mass of the plate? What about the mass of just the “outer” 5 inches?

Example 104. Consider a thin rod oriented on the x -axis over the interval $[\frac{5\pi}{6}, \frac{7\pi}{6}]$, where x is a distance in meters from zero. If the density of the rod is given by the function $\rho(x) = 2\sin(x) + 6$, in kilograms per meter, what is the mass of the rod in kilograms?

Definition 17. The **work** done by a variable force F along a line from $x = a$ to $x = b$ in the direction of the force is:

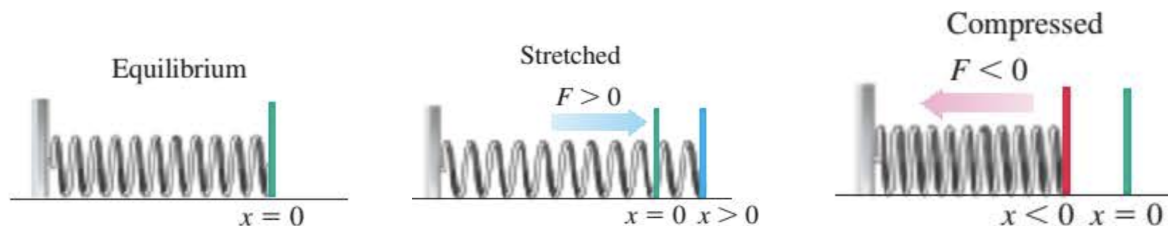
$$W = \int_a^b F(x) dx \quad (\text{in Joules, } J)$$



Observe: We are abstracting “area under the curve” on $[a, b]$, to measure **work**.

Example 105. Work from force. Set up an integral to measure the work required to move an object from $x = 1$ to $x = 3$ (in meters) in the presence of a force (in Newtons, N) given by $F(x) = \frac{2}{x^2}$, acting along the x -axis. The **units for work** are Joules, $J = N \cdot m$.

Example 106. A spring in its equilibrium (on the x -axis) has a force $F(x) = 100x$ applied to it. Set up an integral to measure how much work (in Joules, J) is done to compress the spring $0.5m$.



Example 107. Suppose 20 N of force is required to stretch a spring 0.1 m from its equilibrium.

1. Assuming the spring obeys Hooke's Law, find the function for force.

Definition 18. Hooke's Law: Force required to keep a spring in position x units from equilibrium is: $F(x) = k \cdot x$.

2. Set up an integral to measure the amount of work needed to stretch the spring from $x = 0$ to $x = 0.25$.
3. Set up an integral to measure the amount **additional work** required to stretch the spring from $x = 0.25$ to $x = 3$ from the equilibrium.

Example 108. A spring requires 300 J to stretch from 2 m to 8 m . Assuming the spring obeys Hooke's law, find the spring constant k . Then determine the function that models the force required to stretch/compress the spring from equilibrium.

Section 2.5b: Physical Applications - Pumping, Hydrostatic Force

Definition 19.

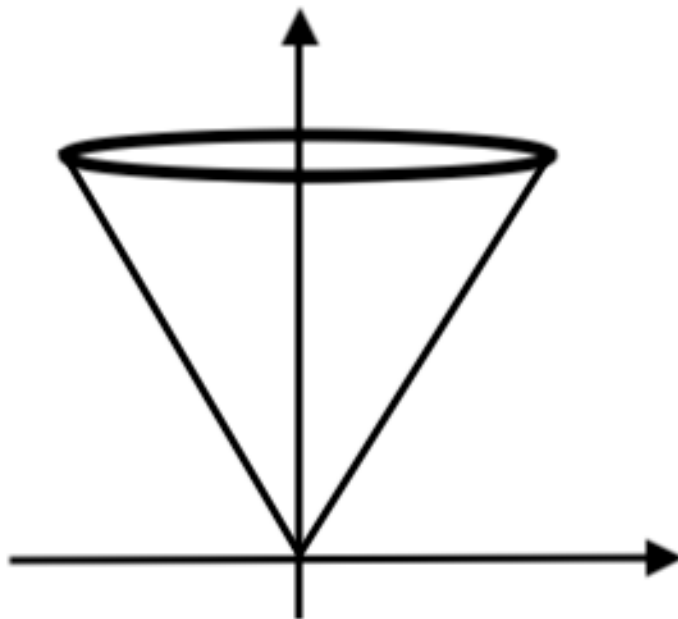
$$\text{Work} = (\text{Force}) \cdot (\text{Distance})$$

Vertical Pumping: “Split into pieces”

$$W_k = F_k \cdot D_k = \left(\rho \cdot V_k \cdot g \right) \cdot y_k \quad \implies$$

Example 109. For a given conical tank, the radius is 7 m and the height is 10 m . Acceleration due to gravity is $9.8 \frac{\text{m}}{\text{sec}^2}$, and the density of water is $1000 \frac{\text{kg}}{\text{m}^3}$.

1. Set up the integral needed to determine how much work is required to pump all of the water over the top edge of the tank.

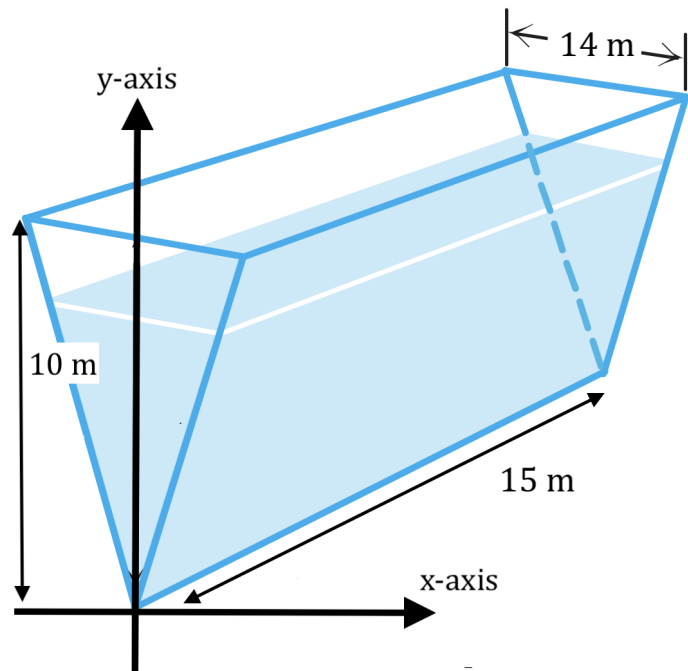


2. Set up the integral needed to determine how much work is required to pump the top 4 m of water out over the top edge of the tank. **Reflect** on modifications needed in the integral set up.

Example 110. For a given olive oil storage tank with triangular ends, the width is 14 m , the height is 10 m , and the length is 15 m . The tank is filled to a depth of 8 m . Acceleration due to gravity is $9.8 \frac{\text{m}}{\text{sec}^2}$, and the density of olive oil is $900 \frac{\text{kg}}{\text{m}^3}$.

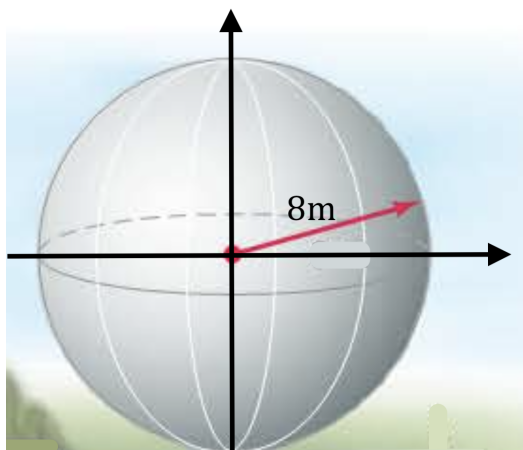
1. Set up the integral needed to determine how much work is required to pump the top 4 m of olive oil out through a standpipe that is 3 m above the top of the tank.

Reflect on the modifications (due to geometry) needed in the integral set up from the previous example.

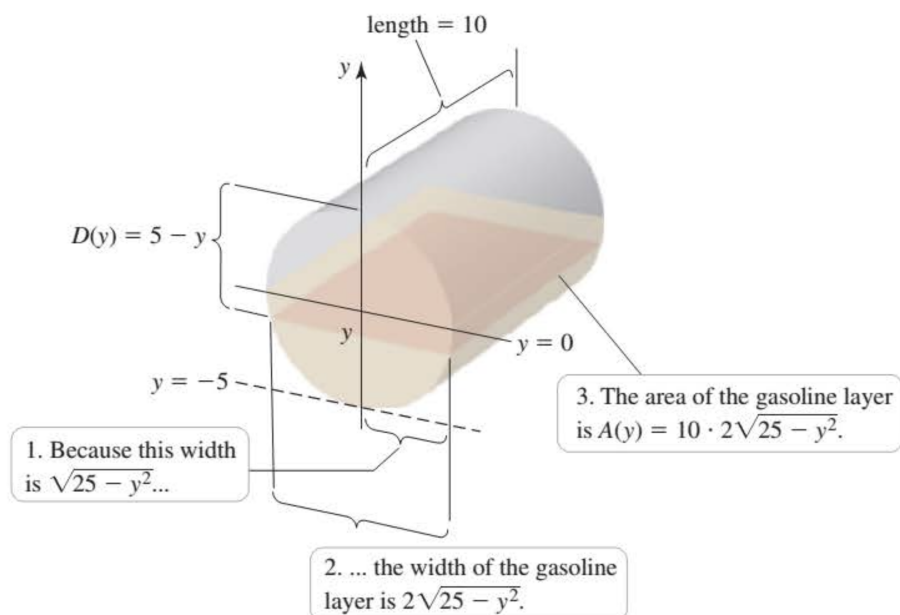


2. Set up the integral needed to determine how much work is required to pump the olive oil out through a standpipe that is 3 m above the top of the tank until 1 m depth of olive oil remains. **Reflect** on changes needed in the integral structure.

Example 111. A spherical tank with a radius of 8 m is completely full of sea water. The sea water is being pumped out the the very top of the tank. Acceleration due to gravity is $9.8 \frac{\text{m}}{\text{sec}^2}$, and the density of (salty) sea water is $1030 \frac{\text{kg}}{\text{m}^3}$. Set up the integral needed to determine how much work is required to pump all the sea water out through the very top. **Discuss** strategies for evaluating this integral.

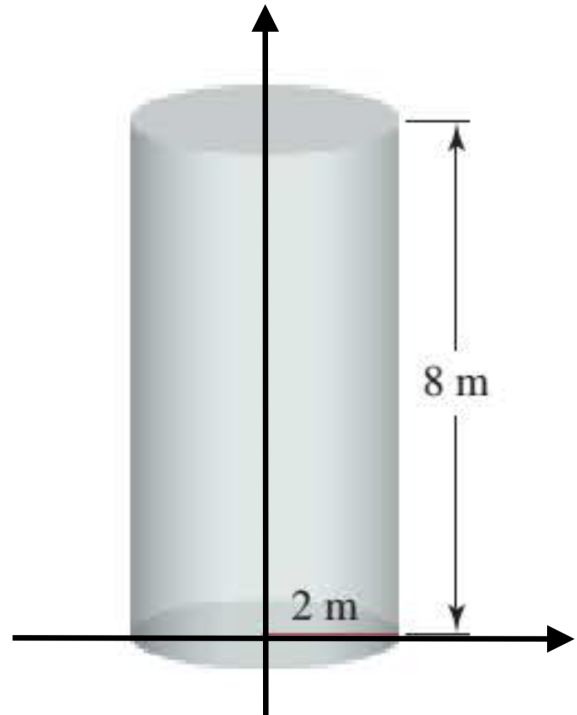


Example 112. Pumping gasoline out of a cylindrical tank on its side with a length of 10 m and a radius of 5 m half full of gasoline. **Discuss** the integral set up for the work require to empty the tank through an outlet pipe at the top of the tank. Density of gasoline is $\rho = 737 \text{ kg/m}^3$.



Example 113. Set up an integral to measure the work needed to pump all the water out of a cylindrical tank with height 8 m , radius 2 m , and outflow 15 m above the bottom of the tank.

1. With a completely filled tank.



2. With a half-full tank.

Hydrostatic Force

- **Horizontal plate:** $F = (\text{volume} \cdot \text{density}) \cdot g = \rho \cdot g \cdot A \cdot D,$

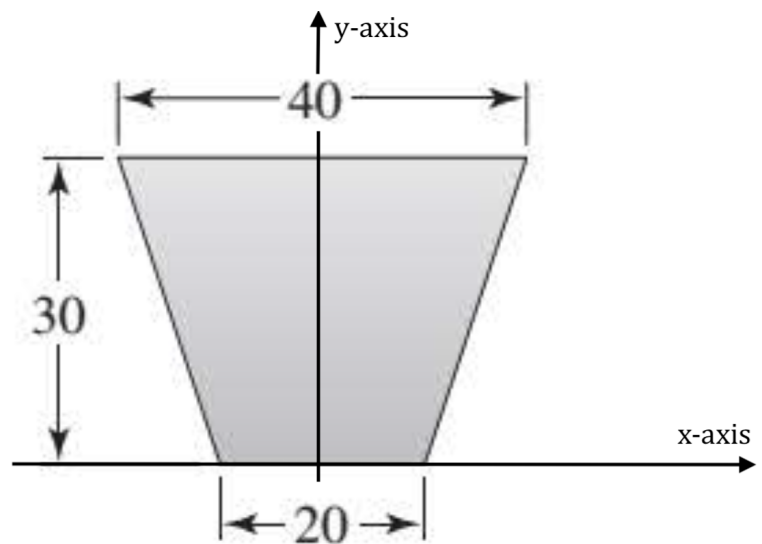
for A area, D vertical depth under the top of the fluid, and ρ mass density (so $\rho \cdot g$ is weight density).

- **Vertical plate:** “Split into pieces.”

$$F_k = \rho \cdot g \cdot A_k \cdot D_k = \rho \cdot g \cdot (W(y_k) \Delta y) \cdot D_k = (\rho \cdot g \cdot V_k) \cdot D_k \quad \Rightarrow$$

Example 114. A trapezoidal faced dam is 30 m in height, has a 20 m base, and 40 m wide top. Set up an integral to measure the total force on the face of the dam.

1. With the dam completely full.

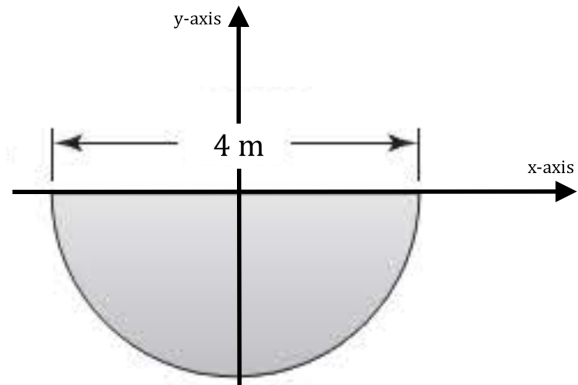


2. With the dam halfway full.

Example 115. Force on a window. An aquarium pool that is 10 m deep and full of water has a semicircular viewing window on one side of its vertical wall. The semicircular window has a 2 m radius. Set up the integral needed to measure the total force acting on the viewing window from the water if...

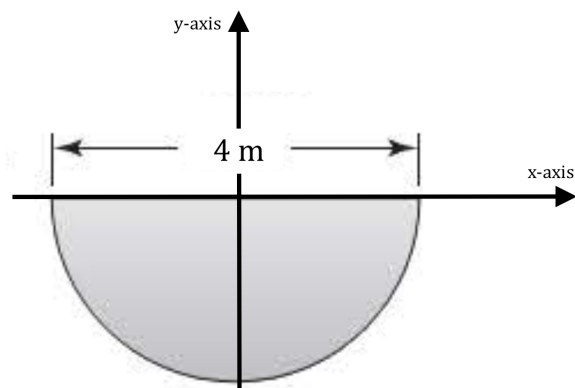
1. The top of the water goes up to the top of the viewing window.

Discuss strategies evaluating the integral.



2. The top of the window is submerged 3 m below the top of the water.

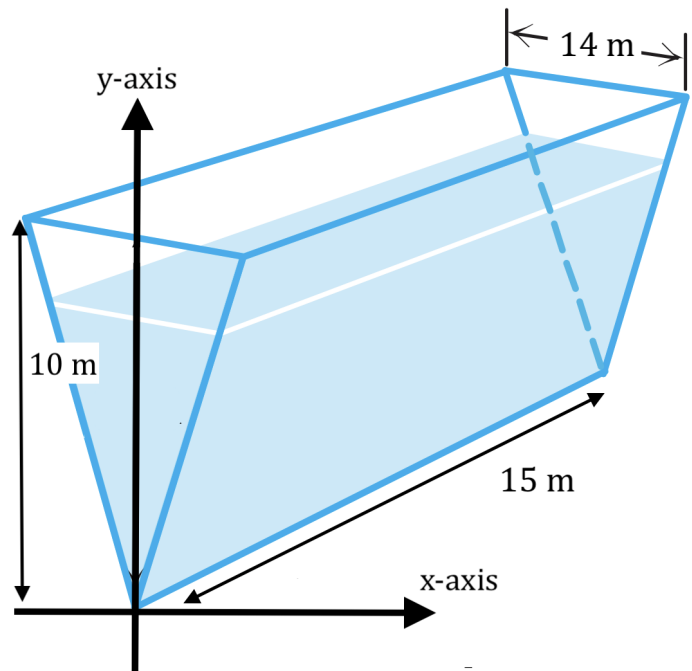
Discuss strategies evaluating the integral.



Example 116. For a given olive-oil storage tank with triangular ends, the width is 14 m , the height is 10 m , and the length is 15 m . The tank is filled with olive oil to a depth of 8 m . Acceleration due to gravity is $9.8\frac{\text{m}}{\text{sec}^2}$, and the density of olive oil is $900\frac{\text{kg}}{\text{m}^3}$.

1. Set up an integral to measure the total force on one triangular end of the tank.

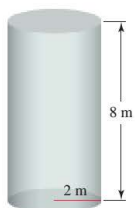
Reflect on which pieces of information were needed to complete this problem.



2. Using the previous answer, set up an integral(s) to measure the total force on both triangular ends of the tank.

15. $\rho(x) = 2 - \frac{x}{2}$, for $0 \leq x \leq 2$
16. $\rho(x) = 1 + 3 \sin x$, for $0 \leq x \leq \pi$
17. $\rho(x) = x\sqrt{2 - x^2}$, for $0 \leq x \leq 1$
18. $\rho(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 2 \\ 2 & \text{if } 2 < x \leq 3 \end{cases}$
19. $\rho(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 2 \\ 1 + x & \text{if } 2 < x \leq 4 \end{cases}$
20. $\rho(x) = \begin{cases} x^2 & \text{if } 0 \leq x \leq 1 \\ x(2 - x) & \text{if } 1 < x \leq 2 \end{cases}$
21. **Work from force** How much work is required to move an object from $x = 0$ to $x = 3$ (measured in meters) in the presence of a force (in N) given by $F(x) = 2x$ acting along the x -axis?
22. **Work from force** How much work is required to move an object from $x = 1$ to $x = 3$ (measured in meters) in the presence of a force (in N) given by $F(x) = \frac{2}{x^2}$ acting along the x -axis?
23. **Compressing and stretching a spring** Suppose a force of 30 N is required to stretch and hold a spring 0.2 m from its equilibrium position.
 - a. Assuming the spring obeys Hooke's law, find the spring constant k .
 - b. How much work is required to compress the spring 0.4 m from its equilibrium position?
 - c. How much work is required to stretch the spring 0.3 m from its equilibrium position?
 - d. How much additional work is required to stretch the spring 0.2 m if it has already been stretched 0.2 m from its equilibrium position?
24. **Compressing and stretching a spring** Suppose a force of 15 N is required to stretch and hold a spring 0.25 m from its equilibrium position.
 - a. Assuming the spring obeys Hooke's law, find the spring constant k .
 - b. How much work is required to compress the spring 0.2 m from its equilibrium position?
 - c. How much additional work is required to stretch the spring 0.3 m if it has already been stretched 0.25 m from its equilibrium position?
25. **Work done by a spring** A spring on a horizontal surface can be stretched and held 0.5 m from its equilibrium position with a force of 50 N.
 - a. How much work is done in stretching the spring 1.5 m from its equilibrium position?
 - b. How much work is done in compressing the spring 0.5 m from its equilibrium position?
26. **Shock absorber** A heavy-duty shock absorber is compressed 2 cm from its equilibrium position by a mass of 500 kg. How much work is required to compress the shock absorber 4 cm from its equilibrium position? (A mass of 500 kg exerts a force (in newtons) of 500g, where $g = 9.8 \text{ m/s}^2$.)
27. **Calculating work for different springs** Calculate the work required to stretch the following springs 0.5 m from their equilibrium positions. Assume Hooke's law is obeyed.
 - a. A spring that requires a force of 50 N to be stretched 0.2 m from its equilibrium position
 - b. A spring that requires 50 J of work to be stretched 0.2 m from its equilibrium position
28. **Calculating work for different springs** Calculate the work required to stretch the following springs 0.4 m from their equilibrium positions. Assume Hooke's law is obeyed.
 - a. A spring that requires a force of 50 N to be stretched 0.1 m from its equilibrium position
 - b. A spring that requires 2 J of work to be stretched 0.1 m from its equilibrium position
29. **Calculating work for different springs** Calculate the work required to stretch the following springs 1.25 m from their equilibrium positions. Assume Hooke's law is obeyed.
 - a. A spring that requires 100 J of work to be stretched 0.5 m from its equilibrium position
 - b. A spring that requires a force of 250 N to be stretched 0.5 m from its equilibrium position
30. **Work function** A spring has a restoring force given by $F(x) = 25x$. Let $W(x)$ be the work required to stretch the spring from its equilibrium position ($x = 0$) to a variable distance x . Find and graph the work function. Compare the work required to stretch the spring x units from equilibrium to the work required to compress the spring x units from equilibrium.
31. **Winding a chain** A 30-m-long chain hangs vertically from a cylinder attached to a winch. Assume there is no friction in the system and the chain has a density of 5 kg/m.
 - a. How much work is required to wind the entire chain onto the cylinder using the winch?
 - b. How much work is required to wind the chain onto the cylinder if a 50-kg block is attached to the end of the chain?
32. **Coiling a rope** A 60-m-long, 9.4-mm-diameter rope hangs freely from a ledge. The density of the rope is 55 g/m. How much work is needed to pull the entire rope to the ledge?
33. **Winding part of a chain** A 20-m-long, 50-kg chain hangs vertically from a cylinder attached to a winch. How much work is needed to wind the upper half of the chain onto the winch?
34. **Leaky bucket** A 1-kg bucket resting on the ground contains 3 kg of water. How much work is required to raise the bucket vertically a distance of 10 m if water leaks out of the bucket at a constant rate of $\frac{1}{3}$ kg/m? Assume the weight of the rope used to raise the bucket is negligible. (*Hint:* Use the definition of work, $W = \int_a^b F(y) dy$, where F is the variable force required to lift an object along a vertical line from $y = a$ to $y = b$.)
35. **Emptying a swimming pool** A swimming pool has the shape of a box with a base that measures 25 m by 15 m and a uniform depth of 2.5 m. How much work is required to pump the water out of the pool (to the level of the top of the pool) when it is full?
36. **Emptying a cylindrical tank** A cylindrical water tank has height 8 m and radius 2 m (see figure).
 - a. If the tank is full of water, how much work is required to pump the water to the level of the top of the tank and out of the tank?

- b. Is it true that it takes half as much work to pump the water out of the tank when it is half full as when it is full? Explain.

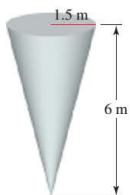


37. **Emptying a half-full cylindrical tank** Suppose the water tank in Exercise 36 is half full of water. Determine the work required to empty the tank by pumping the water to a level 2 m above the top of the tank.

38. **Emptying a partially filled swimming pool** If the water in the swimming pool in Exercise 35 is 2 m deep, then how much work is required to pump all the water to a level 3 m above the bottom of the pool?

39. **Emptying a conical tank** A water tank is shaped like an inverted cone with height 6 m and base radius 1.5 m (see figure).

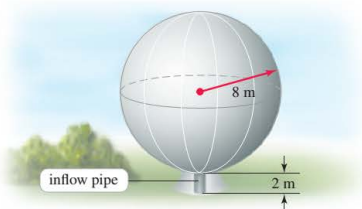
- a. If the tank is full, how much work is required to pump the water to the level of the top of the tank and out of the tank?
b. Is it true that it takes half as much work to pump the water out of the tank when it is filled to half its depth as when it is full? Explain.



40. **Upper and lower half** A cylinder with height 8 m and radius 3 m is filled with water and must be emptied through an outlet pipe 2 m above the top of the cylinder.

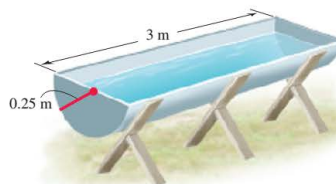
- a. Compute the work required to empty the water in the top half of the tank.
b. Compute the work required to empty the (equal amount of) water in the lower half of the tank.
c. Interpret the results of parts (a) and (b).

41. **Filling a spherical tank** A spherical water tank with an inner radius of 8 m has its lowest point 2 m above the ground. It is filled by a pipe that feeds the tank at its lowest point (see figure). Neglecting the volume of the inflow pipe, how much work is required to fill the tank if it is initially empty?



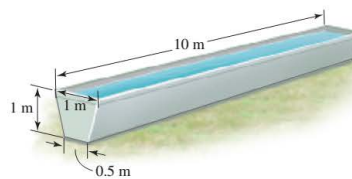
42. **Emptying a water trough** A water trough has a semicircular cross section with a radius of 0.25 m and a length of 3 m (see figure).

- a. How much work is required to pump the water out of the trough (to the level of the top of the trough) when it is full?
b. If the length is doubled, is the required work doubled? Explain.
c. If the radius is doubled, is the required work doubled? Explain.



43. **Emptying a water trough** A cattle trough has a trapezoidal cross section with a height of 1 m and horizontal sides of length $\frac{1}{2}$ m and 1 m. Assume the length of the trough is 10 m (see figure).

- a. How much work is required to pump the water out of the trough (to the level of the top of the trough) when it is full?
b. If the length is doubled, is the required work doubled? Explain.



44. **Pumping water** Suppose the tank in Example 5 is full of water (rather than half full of gasoline). Determine the work required to pump all the water to an outlet pipe 15 m above the bottom of the tank.

45. **Emptying a conical tank** An inverted cone (base above the vertex) is 2 m high and has a base radius of $\frac{1}{2}$ m. If the tank is full, how much work is required to pump the water to a level 1 m above the top of the tank?

Section 3.7: Improper Integrals

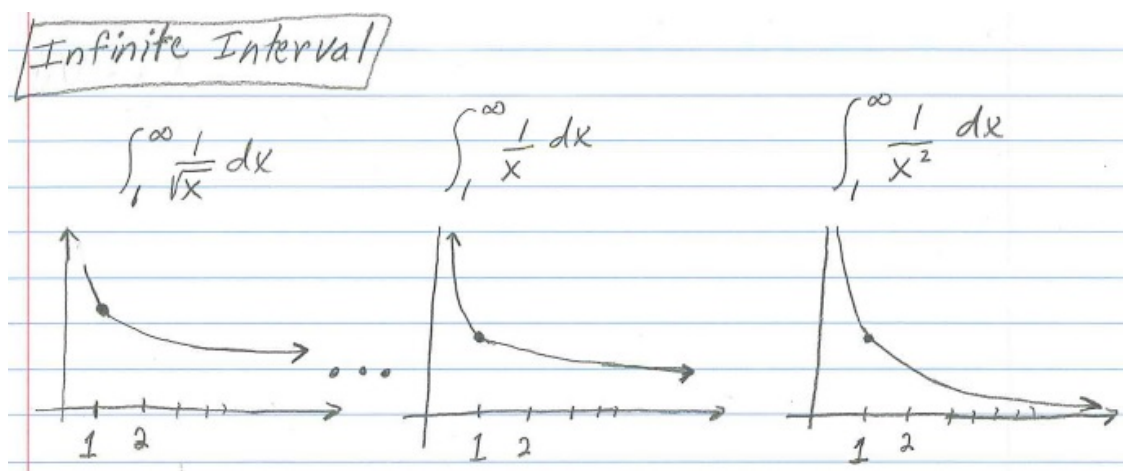
Definition 20. An integral is *improper* if either ...

Type I. A bound of integration is infinite, e.g.

$$\int_1^{\infty} \frac{1}{x^2} dx$$

Type II. The function is undefined (or “infinite”) at an endpoint, e.g.

$$\int_0^1 \frac{1}{x} dx$$



Definition 21. For a continuous function f ,

$$\int_a^{\infty} f(x) dx = \lim_{N \rightarrow \infty} \int_a^N f(x) dx.$$

The integral **converges** if the limit converges to a finite value. Otherwise, the integral is **divergent**.

Example 117. Evaluate the improper integral to determine if it converges or diverges.

$$\int_1^{\infty} \frac{1}{x^2} dx$$

Example 118. Evaluate the improper integral to determine if it converges or diverges.

$$\int_0^{\infty} \frac{2}{(x+1)^3} dx$$

Example 119. Evaluate the improper integral to determine if it converges or diverges.

$$\int_0^{\infty} \frac{1}{\sqrt{x+1}} dx$$

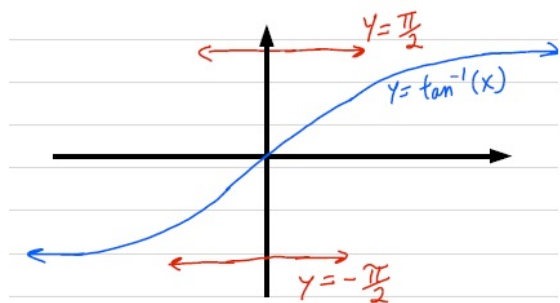
Observe how the exponent affects convergence.

Example 120. Evaluate the improper integral to determine if it converges or diverges.

$$\int_0^{\infty} e^{-x} dx$$

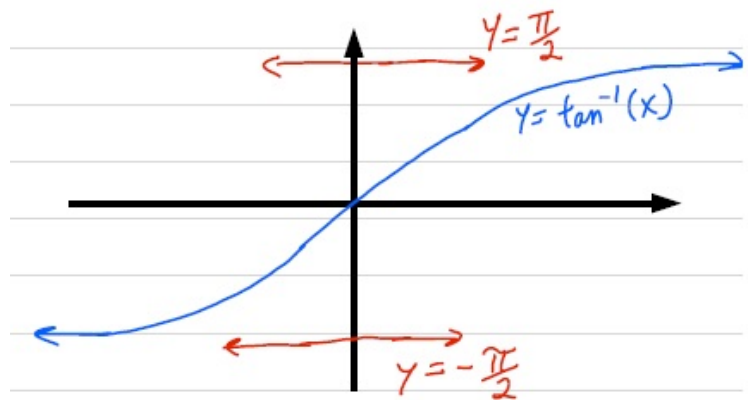
Example 121. Evaluate the improper integral to determine if it converges or diverges. If the integral is convergent, keep your answer in exact form in terms of π .

$$\int_{-\infty}^7 \frac{2}{x^2 + 49} dx$$

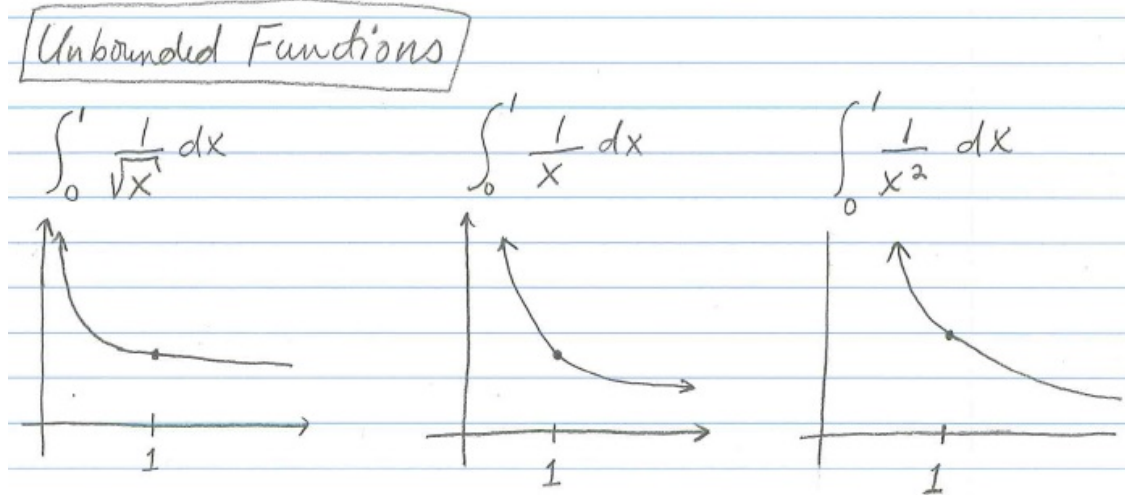


Example 122. Evaluate the improper integral to determine if it converges or diverges. If the integral is convergent, keep your answer in exact form in terms of π .

$$\int_{-\infty}^{\infty} -\frac{4}{x^2 + 25} dx$$



Type II. Improper Integrals.



Definition 22. For f continuous on $(a, b]$,

$$\int_a^b f(x) dx = \lim_{A \rightarrow a^+} \int_A^b f(x) dx$$

Example 123. Evaluate the improper integral to determine if it converges or diverges.

$$\int_0^1 \frac{1}{\sqrt{x}} dx$$

Example 124. Evaluate the improper integral to determine if it converges or diverges.

$$\int_{-2}^1 \frac{1}{\sqrt{x+2}} dx$$

Example 125. Evaluate the improper integral to determine if it converges or diverges.

$$\int_1^7 \frac{1}{x-1} dx$$

Definition 23. For f continuous on $[a, c) \cup (c, b]$,

$$\int_a^b f(x) \, dx = \lim_{A \rightarrow c^+} \int_A^b f(x) \, dx + \lim_{B \rightarrow c^-} \int_a^B f(x) \, dx$$

Example 126. Evaluate the improper integral to determine if it converges or diverges.

$$\int_{-5}^2 \frac{1}{x^2} \, dx$$

Example 127. Evaluate the improper integral to determine if it converges or diverges.

$$\int_{3\pi/8}^{\pi/2} 2 \sec^2(4x) \, dx$$

Example 128. Evaluate the integral

$$\int_{-\infty}^{\infty} \frac{4}{x^2 + 25} dx$$

If the integral is divergent, enter $\emptyset$.